



Double-layer optimal scheduling for wind-PV-hydro-hybrid energy storage system with multi-timescale coordination

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ABSTRACT

A multi-scenario coordinated control method for wind-photovoltaic-hydro-hybrid energy storage system is proposed to address the challenges of intensified power fluctuations and underutilized energy storage resulting from high-penetration renewable integration. Typical wind and photovoltaic scenarios are generated using Latin Hypercube Sampling and reduced via an improved Load Curve-ISODATA clustering algorithm to characterize output uncertainty. A mixed-integer programming model is developed to minimize system output variability, with pumped storage utilized for inter-day energy shifting and battery energy storage for minute-level fluctuation suppression. Power allocation under hydropower-storage coupling constraints is optimized using the Gurobi solver. A capacity allocation model is formulated based on the Competition of Tribes and Cooperation of Members (CTCM) algorithm to enhance ancillary service revenues. In this framework, pumped storage prioritizes peak-valley arbitrage, while battery energy storage is allocated to fast-response frequency regulation. The effectiveness of the method proposed in this paper was verified through simulation of actual operating data from a northwest power grid in China. Calculation results indicate that the method proposed in this paper can reduce system volatility by 12.46 % while increasing revenue by \$34,762. The system's return on investment reaches 15.3 %, with a payback period of 6.1 years, effectively enhancing both system stability and economic efficiency.

1. Introduction

With the deepening implementation of the "Dual-Carbon" strategy, the installed capacity of renewable energy sources (RES), represented by wind and photovoltaic power, has experienced exponential growth. However, the large-scale integration of highly volatile wind and photovoltaic power into the grid poses significant challenges to the reliability and stability of power systems [1,2]. Hydropower, through the rapid operation of turbine units, provides large-scale adjustable electricity [3,4]. Building upon this, hydropower units and pumped storage units can effectively mitigate the variability and uncertainty of wind-photovoltaic power outputs. Electrochemical energy storage, renowned for its rapid response and high energy conversion efficiency, significantly curtails RES-induced fluctuations [5,6]. Consequently, constructing a coordinated wind-photovoltaic-hydro-hybrid energy storage system (WPHHS) has become a critical pathway for enhancing the resilience of modern power systems.

To address the high proportion of new energy integration, recent studies have proposed various methods to optimize the integration and

dispatch of renewable energy. However, most of these studies exhibit certain shortcomings that require further refinement. *Hedayatnia A et al.* [7] proposed a microgrid energy management method based on deep reinforcement learning. Although this approach achieved good results in real-time dynamic scheduling of microgrids, it primarily focused on short-term energy optimization. It lacked consideration for comprehensive scheduling across multiple time scales and energy sources, as well as long-term optimization, limiting its application in large-scale power grids. *Sepehrzad R et al.* [8] proposed an intelligent dynamic energy management approach based on demand response and energy storage systems. While this research has made some progress in the dispatch of smart city microgrids, it fails to thoroughly address the coordinated dispatch of different energy forms within multi-energy complementary systems. Particularly when confronting the fluctuations of wind and solar power, the system's stability and economic viability remain difficult to guarantee. *Sepehrzad R et al.* [9] focus on energy management within integrated energy distribution systems, considering interactions in energy markets and end-user participation. Their approach effectively manages the variability of different energy

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sources through short-term probabilistic scenario optimization. However, it primarily concentrates on the interaction between electricity markets and demand response, failing to fully account for the role of energy storage systems in mitigating variability. Furthermore, it lacks in-depth research on the coordinated optimization of multi-energy systems. The above studies reveal that while significant progress has been made in microgrids, demand response, and energy management, these approaches generally suffer from inadequate coordination within multi-energy systems. Most research focuses on optimizing the scheduling of single energy sources or technologies, failing to fully consider the synergistic scheduling and optimization of multiple energy forms [10–12].

Numerous scholars worldwide have achieved notable progress in multi-energy complementary optimal scheduling for wind-photovoltaic-hydro system. **Ariyaratna PM et al.** [13] developed a cross-regional wind-photovoltaic-hydro complementary system optimization model, demonstrating hydropower's effectiveness in stabilizing RES fluctuations, yet overlooked the impact of wind-photovoltaic output uncertainty under extreme weather conditions on scheduling outcomes. **Zhao H et al.** [14] employed a multi-agent system to address short-term wind-photovoltaic-hydro scheduling modeling and optimization, offering decision-makers flexible trade-offs between operational efficiency and costs. However, it failed to account for the high volatility and unpredictability of wind and photovoltaic power, limiting its ability to manage operational risks under strong RES uncertainty. **Jin X et al.** [15] proposed the SDDP framework to address the long-term complementary dispatch of hydro-wind-solar systems, integrating multiple uncertainties with year-end reservoir capacity management. This approach increased channel utilization rates by over 80 %, but its failure to account for short-term volatility limited its fine-tuning capabilities. **Tapia Carpio LG et al.** [16] proposed a mean-variance model to optimize regional combinations of hydro, wind, and solar power in Brazil. By constructing an efficient frontier, they achieved smoother output fluctuations, revealing that the current wind power mix is efficient but highly volatile, solar power lies far from the frontier, and hydro power is efficient with low volatility. However, the model is based solely on resource input variability and does not incorporate practical operational conditions such as grid transmission capacity and dispatch constraints. **Liu B et al.** [17] proposed an opportunity-constrained cascade section nesting model that leverages hydropower flexibility to coordinate wind and solar integration with transmission bottlenecks. This approach significantly reduced curtailment rates for wind and solar power (99.1 % during the flood season) and smoothed peak-to-valley load differences (54.4 % reduction). However, it did not extend to multi-stakeholder economic analysis. The above studies are highly dependent on the flexible regulation capability of hydropower [18], but multiple uncertainties, extreme weather and forecast bias can significantly limit its regulation effect, leading to reduced unit efficiency [19], increased wind and photovoltaic power abandonment rates and reduced system reliability [20]. Battery energy storage can decouple time rigidity and shift fluctuating energy to peak demand; overcome spatial constraints and hedge cross-regional errors [21]; protect core equipment and reduce frequent hydropower regulation. The rational use of energy storage in multi-energy systems can effectively improve the above problems.

The multi-energy complementary system with storage has also made great achievements in the research on optimal scheduling. **Wang H et al.** [22] proposes a short-term joint dispatch model for wind-solar-hydro-storage systems with variable-speed pumped storage units, following the "variable speed unit + minimum start-stop principle" strategy to optimize system fluctuation control and cost efficiency at the same time. This study combines the characteristics of complex pumping units to enhance the adaptability of the model to the actual operation of the power grid. However, the model has a large amount of calculation when solving, high algorithm complexity, and strong dependence on data accuracy and parameters. **Li Z et al.** [23] proposes a mid-term (multi-stage) distributed robust optimization model, focusing

on price arbitrage and uncertainty management of wind solar water storage systems in the mid-term market, which is more suitable for the mid-term market environment and has good robustness; However, the model structure is complex, with numerous parameters, and relies on refined system modeling and market data. The feasibility of real-time scheduling or multi zone system expansion still needs improvement. **Zhang Q et al.** [24] proposes a two-stage dispatch optimization model for wind-solar-hydro-pumped storage multi-energy system. Its main advantages are clear modeling structure, strong practicality of dispatch strategy, and the improvement of renewable energy utilization rate verified by examples. However, the model does not take into account practical factors such as energy storage response characteristics, energy efficiency loss and capacity degradation, which limits its application and promotion under more complex working conditions. **Boretti A et al.** [25] quantified that Australia's pure wind-solar grid would require 327 GW of wind-solar capacity plus 48 TWh of hydrogen storage (electrolyzer peak capacity 267GW/average capacity 46 GW), a result consistent with the Royal Society's projections. However, they did not explore nuclear power as an alternative for uranium-rich Australia, nor did they conduct economic analyses involving multiple investors. **Geng C et al.** [26] propose a PVSyst-TRNSYS coupled model to optimize hybrid wind-solar-geothermal systems for high-altitude off-grid buildings, achieving a 95 % renewable energy share and an electricity cost of \$0.12/kWh. However, the study fails to quantify carbon emission reductions, and the economic model does not account for fluctuations in policy subsidies. **Jing Z et al.** [27] A et al. proposed a cascade hydro-wind-solar-storage complementary dispatch framework based on a multi-dimensional three-dimensional power generation database. They innovatively quantified that reversible pumped hydro storage increases renewable energy absorption rates by 40.78 %. However, this approach fails to integrate the disruptive impact of climate change on runoff and relies on simulated wind and solar data. In the above studies, the energy storage system is limited to a single application scenario, resulting in low energy storage utilization and insufficient economy.

Regarding multi-scenario applications of energy storage, researchers have made incremental advancements. **Cao Z et al.** [28] proposed a dual-objective optimization model based on weight coefficient λ transformation. By solving it with PSO, they obtained the optimal energy storage configuration (80kW/625 kWh reducing wind power fluctuation to 18.6 %). However, they did not analyze the sensitivity impact of λ , and the economic model did not incorporate practical factors such as policy subsidies. **Dicorato M et al.** [29] proposed a market participation strategy for wind-storage systems based on dynamic programming, demonstrating that 15 MW of energy storage can improve schedule tracking rates and respond to electricity price fluctuations. However, the study did not quantify battery degradation costs, and its economic model relies on Italy's specific subsidy policies. **Han X et al.** [30] proposed a multi-scenario shared control method for wind-solar hybrid pumped storage, integrating load tracking and ancillary service optimization to achieve load deviation < 2 %. However, this approach does not incorporate medium-to-long-term hydrological forecasts, and its economic model relies on region-specific electricity pricing policies. **Han X et al.** [31] proposed a dual-layer dynamic game optimization method to optimize capacity allocation and operational strategies in multi-scenario shared energy storage systems, thereby increasing annual revenue and high-value peak shaving ratios. However, the model suffers from high computational complexity, simulation dependency on specific data, and insufficient validation of its generalization across all seasonal variations. The results show that the annual total income and high-value peak-shaving rate are the highest under multiple scenarios, and the cost-to-income ratio is at an acceptable low level. However, these works focused solely on single-type storage system, lacking the adaptability required for complex multi-modal scenarios [32].

When solving multi-energy complementary systems containing clean RES, researchers usually use optimization algorithms. **Jamal R et al.** [33] proposed an improved artificial hummingbird algorithm for the

stochastic optimal reactive power dispatch problem containing renewable energy sources that avoids local optimality by enlisting flight and distance bandwidth motion, combined with Monte Carlo simulation to deal with wind and water uncertainty. The results show significant reduction of grid losses, improvement of voltage profile and stability, and better performance than mainstream optimization algorithms. **Modu B et al.** [34] proposed a framework for the efficient design and evaluation of independent hybrid renewable energy systems (HRES) to meet the energy needs of rural communities in northeastern Nigeria, with the goal of minimizing the annual system cost of HRES. At the same time, the reliability constraints of the probabilistic loss of power supply (LPSP) was also considered. Compared with multiple algorithms, LF-SSA has the best solution. **Wang Z et al.** [35] proposed a new multi-objective optimization framework for risk-managed economic dispatch in hybrid hydrothermal-wind-photovoltaic systems (HTWPS). The multi-objective artificial rabbit optimization (MOARO) algorithm was used, combined with the Pareto criterion and fuzzy theory to optimize dispatch decisions while balancing costs and risks. **Wang Y et al.** [36] proposed the FORCL-LSHADE improved algorithm for the multi-region dynamic economic dispatch problem exacerbated by wind/light fluctuations by means of fractional order mutation with a fine learning strategy to optimize uncertainty modeling (based on Weibull/log-normal distribution). **Han X et al.** [37] addresses the problems of uncertainty in renewable energy generation and the single operation mode of energy storage system, and proposes a cooperative game robust optimization control method for a wind-photovoltaic hybrid energy storage system based on the dual settlement model of the electricity market. Simulation results show that the cooperative game robust optimization model achieves the optimal operation of the wind-photovoltaic hybrid energy storage system considering multiple uncertainties, which can improve the system's ability to cope with uncertainty risks and the system's reliability. The algorithms employed in the above papers emphasize efficient exploration-exploitation balance, handling renewable energy uncertainty, and multi-objective optimization. However, they suffer from high computational complexity, parameter sensitivity, slow convergence, or local optimum issues, which limit their large-scale practical application.

The primary contribution of this paper lies in the proposal of a novel multi-scenario coordinated control method that deeply optimizes the volatility and uncertainty of wind, solar, and hydroelectric hybrid energy storage systems. Compared to existing research, this paper's innovations are primarily reflected in the following aspects:

- 1) Innovative Application of the Load Curve-ISODATA (L-ISODATA) Clustering Algorithm: This paper applies the L-ISODATA clustering algorithm to scenario reduction in multi-energy complementary systems, effectively reducing the computational burden caused by uncertainty and improving optimization efficiency.
- 2) Introduction of the Competition of Tribes and Cooperation of Members Algorithm (CTCM): This paper applies the CTCM algorithm to capacity allocation and scheduling optimization in multi-energy systems for the first time, overcoming the vulnerability of existing methods to local optimal solutions and improving the system's global optimization capabilities.
- 3) Multi-scenario and Multi-timescale Coordinated Scheduling: Unlike existing research, this paper not only considers short-term volatility scheduling but also comprehensively considers optimization requirements across multiple timescales, effectively improving renewable energy utilization and enhancing system stability and economic efficiency.

Through these innovations, the proposed method demonstrates significant advantages in the scheduling and optimization of multi-energy systems, improving system robustness and economic efficiency while also filling gaps in existing research.

The subsequent chapters of this paper are organized as follows:

Section 2 addresses the uncertainty of wind and PV generation by generating typical scenarios using LHS and L-ISODATA. Sections 3 establish the WPHHS that considers multi-scenario applications of optimize operation and ancillary services. Section 4 validates the proposed approach through a case study, and Section 5 summarizes the conclusions and outlines future research directions.

2. Scenario generation and reduction for wind-photovoltaic power

2.1. Wind-photovoltaic scenario generation based on LHS

Based on the probability density distributions of wind and photovoltaic power, a sampling algorithm is employed to generate probabilistic forecast scenarios. LHS, a stratified sampling technique [18], approximates random sampling from multivariate parameter distributions. Compared to other sampling methods, LHS offers higher accuracy, smaller sampling sizes, and superior efficiency, thereby improving the estimation of random variable means and variances while ensuring robustness [38]. The LHS process involves two steps: interval sampling and sample correlation sorting.

Historical wind and photovoltaic power data from Northwest China, which follow log-normal distributions, are used to generate 1,000 scenarios via LHS. These scenarios are illustrated in Figs. 1 and 2, where each colored curve represents a potential wind or photovoltaic power output trajectory.

2.2. Scenario reduction based on L-ISODATA

The Iterative Self-Organizing Data Analysis Algorithm (ISODATA) is an unsupervised dynamic clustering method based on statistical pattern recognition. Similar to the K-means algorithm, ISODATA iteratively determines cluster centers by optimizing sample means [39,40]. The L-ISODATA algorithm, an improved version of the ISODATA clustering algorithm, is first applied to load curve clustering [41]. In this study, it is employed for wind-PV scenario reduction.

The L-ISODATA further adapts the number of clusters through splitting and merging operations based on data characteristics. The scenario values at cluster centers are selected as typical scenarios, while the number of scenarios within each cluster is normalized as their probability relative to the total LHS-generated scenarios. The L-ISODATA is applied to reduce the generated wind and photovoltaic power scenarios. The resulting typical scenarios are illustrated in Figs. 3 and 4.

To validate the effectiveness of the proposed clustering method, the

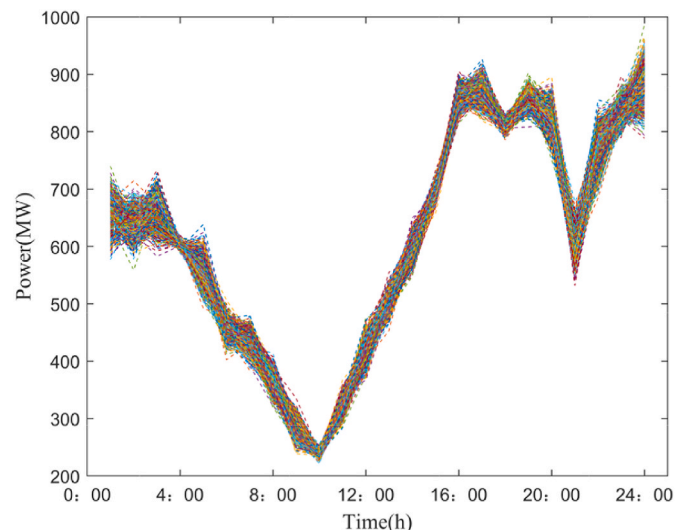


Fig. 1. Wind power output scenarios.

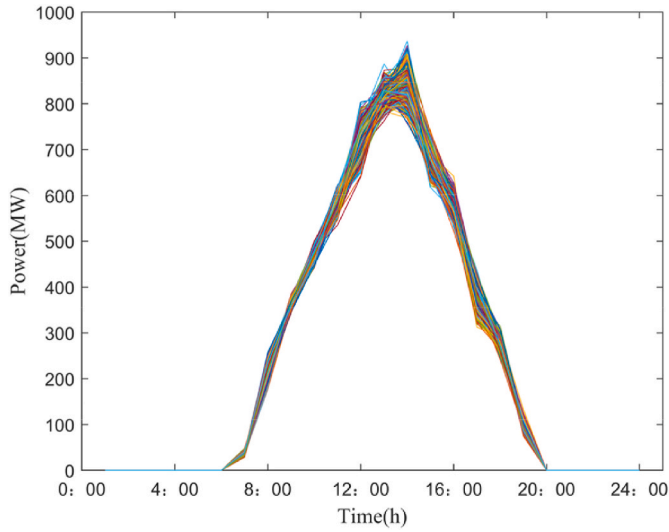


Fig. 2. Photovoltaic power generation output scenarios.

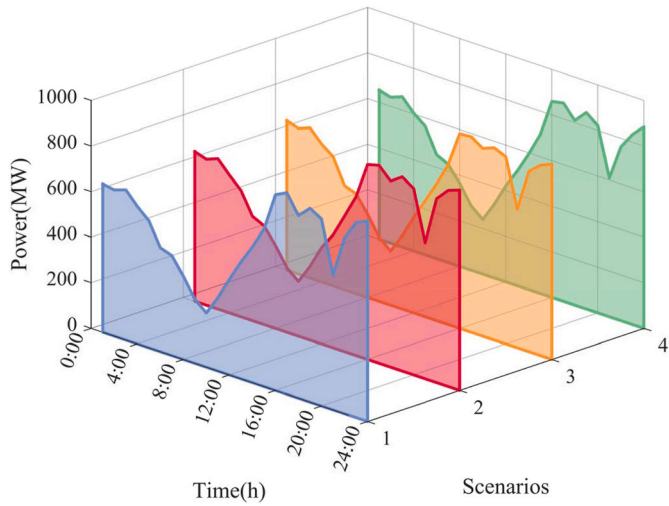


Fig. 3. Typical output scenarios of wind power.

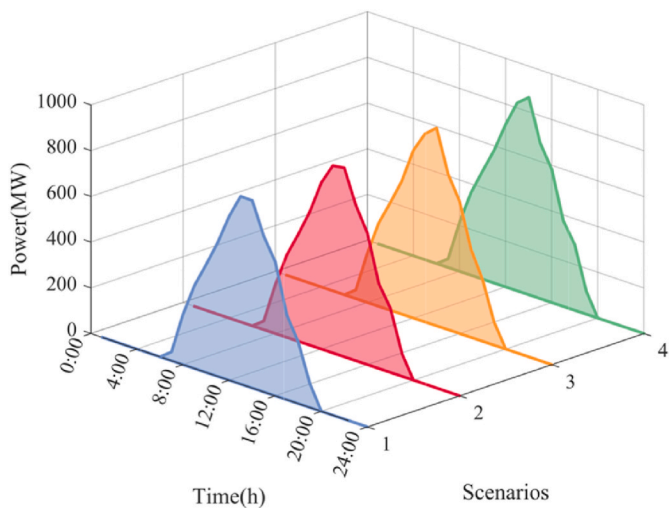


Fig. 4. Typical output scenarios of photovoltaics.

scenario reduction results obtained by L-ISODATA were compared with those of K-means, ISODATA, and K-means-L-ISODATA (K-L-ISODATA). The clustering performance was evaluated using the Davies-Bouldin Index (DBI) and Dunn Index (DI) [42], with comparative results summarized in Table 1.

It can be seen in Table 1, the L-ISODATA scenario reduction algorithm exhibits significant advantages in clustering performance. Taking photovoltaic scenario reduction as an example:

DBI: The L-ISODATA algorithm achieves a DBI value of 0.0066, which is markedly lower than 1.2245 for K-means and 1.1659 for ISO-DATA. This demonstrates that L-ISODATA achieves a superior balance between intra-cluster compactness and inter-cluster separation.

DI: The L-ISODATA algorithm attains the highest DI value of 1.1419, underscoring its exceptional capability in maximizing inter-cluster distinguishability.

Combining the two evaluation indexes of DBI and DI, it can be concluded that the L-ISODATA algorithm is not only effective in reducing the intra-class dispersion, but also has an obvious advantage in maintaining the inter-class differentiation, which fully reflects the excellent performance of the algorithm in clustering tasks.

3. Multi scenario optimization scheduling architecture of the WPHHS

3.1. Optimization scheduling architecture of the WPHHS

WPHHS is mainly composed of wind, photovoltaic generators, hydropower stations, HESS and control center. The Optimization scheduling architecture of the WPHHS is shown in Fig. 5.

It can be seen from Fig. 5, the deep coupling of wind power, photovoltaic power, hydropower and HESS has built an energy system with multi-energy complementarity and coordinated optimization, which is conducive to improving the utilization efficiency of renewable energy and the stability of the power system. As typical intermittent renewable energy sources, wind power and photovoltaic power are significantly affected by meteorological conditions and diurnal changes, showing strong randomness and volatility. In contrast, hydropower units have superior regulation capabilities, which can improve the system's ability to absorb RES to a certain extent. The introduction of energy storage systems can further enhance the system's absorption capacity and enhance coordination and optimization capabilities.

The HESS consists of battery energy storage (BES) and pumped storage (PS). Due to its rapid response characteristics, electric energy storage can effectively suppress short-term fluctuations in wind and photovoltaic power generation. When used in instantaneous power buffering scenario, it can improve the quality of electricity and the stability of grid operation; pumped storage, as a large-capacity, long-cycle energy storage technology, is good at long-term energy transportation. Through the two-way conversion mechanism of "electric energy-potential energy", it absorbs excess electricity when wind and photovoltaic power output is surplus, and releases it at peak load to alleviate the contradiction between supply and demand of the grid. The complementary synergy of the two types of energy storage methods builds a hierarchical, multi-time scale energy storage network, which effectively alleviates the mismatch between the uncertainty of wind and photovoltaic power generation and the load demand of the power grid.

Table 1
Performance comparison of four wind farm scene reduction algorithms.

algorithm	wind		photovoltaic	
	DBI	DI	DBI	DI
K-means	2.2638	0.0023	1.2245	0.0058
ISODATA	2.1928	0.0030	1.1659	0.0070
K-L-ISODATA	0.0060	2.0581	0.0074	1.0829
L-ISODATA	0.0047	2.0856	0.0066	1.1419

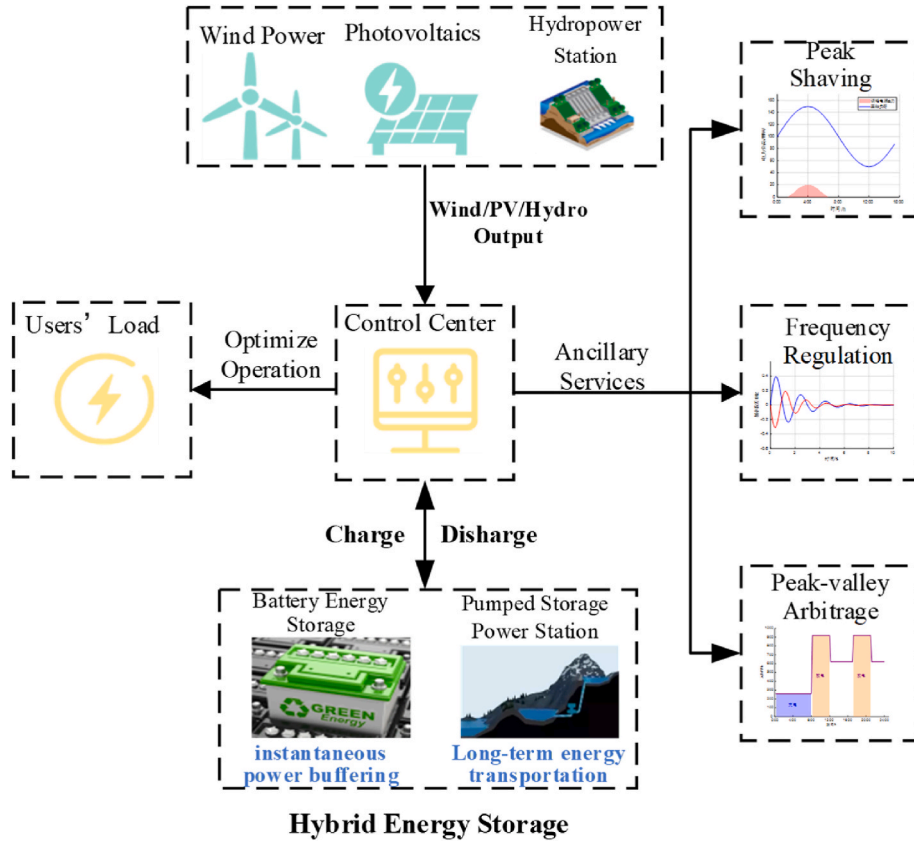


Fig. 5. Optimization scheduling architecture of the WPHHS.

As a new type of hybrid power generation system, the WPHHS organically combines wind energy, photovoltaic energy and hydropower, and utilizes the regulation capacity of hydropower stations and electric energy storage to effectively mitigate system output fluctuations and improve the utilization efficiency of renewable energy. At the same time, the multi-energy complementary system can also be used to participate in ancillary services, fully utilize the system capacity, and increase system benefits.

3.2. Multi-scenario two-layer collaborative optimization strategy

Based on the WPHHS under multiple time scales, a two-layer collaborative optimization strategy including optimized operation and ancillary services is constructed, as shown in Fig. 6.

It can be seen in Fig. 6 that the multi-energy complementary system coordinates diverse energy sources through a control center platform to ensure secure and stable operation. The upper-layer model, targeting minimization of system power fluctuations, addresses the inherent randomness and intermittency of wind and photovoltaic generation by establishing a mixed-integer linear programming (MILP) framework. This framework focuses on the aggregated system output and average power profile, coordinating energy storage devices and conventional generation units to minimize the mean squared deviation of power fluctuations over the operational horizon. The upper-layer model effectively reduces fluctuation amplitudes, enhancing grid stability.

The lower-layer model optimizes economic benefits in ancillary service markets through a CTCM-based remaining capacity allocation strategy, achieving synergistic maximization of revenue from frequency regulation, peak shaving, and peak-valley arbitrage. System revenue originates from two streams:

- 1) Peak-valley arbitrage: Storing energy (e.g., charging storage or pumping water) during low-demand periods with low electricity

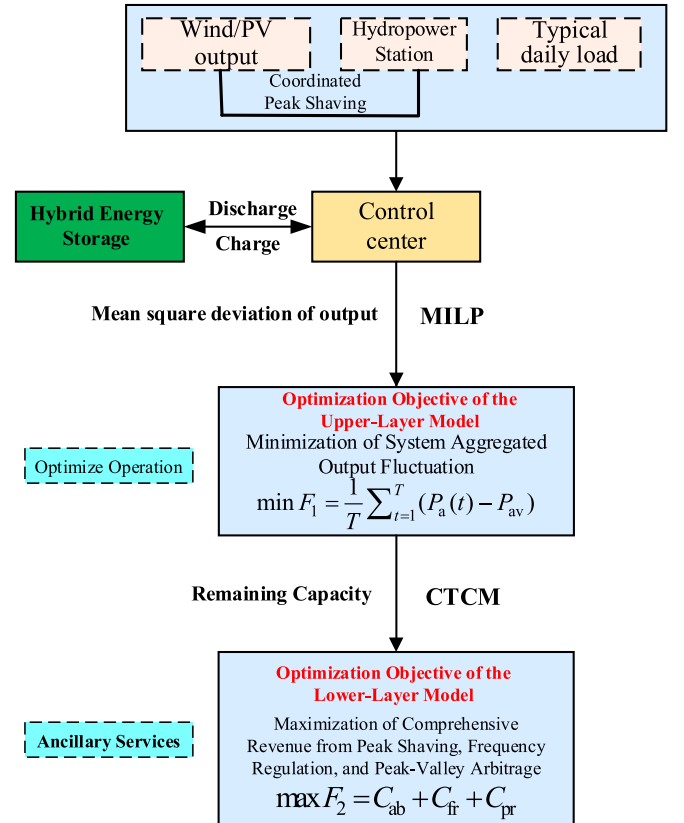


Fig. 6. Dual layer collaborative optimization strategy.

prices and discharging/selling energy during high-demand periods with premium prices.

- 2) Ancillary service compensation: Adjusting power output to maintain grid frequency stability and supply-demand balance, thereby earning service fees from the grid operator.

3.3. Operation optimization model for WPHHS based on MILP

3.3.1. Objective function

The WPHHS primarily integrates PS and BES into the existing wind-photovoltaic-hydro framework. By leveraging the operational flexibility of energy storage, this system mitigates the randomness, intermittency, and volatility in hydro, wind, and photovoltaic power generation caused by geographical and meteorological factors, thereby reducing adverse impacts on grid security and stability. Therefore, an optimization control model is developed with the objective of minimizing system aggregated output fluctuation.

$$\begin{cases} \min F_1 = \frac{1}{T} \sum_{t=1}^T (P_a(t) - P_{av}) \\ P_a(t) = P_w(t) + P_{pv}(t) + P_s(t) + P_{ps}(t) + P_d(t) \\ P_{av} = \frac{1}{T} \sum_{t=1}^T P_a(t) \end{cases} \quad (1)$$

where F_1 is fluctuation index of the system aggregated output; P_{av} is average aggregated output of the WPHHS over one operational cycle; T is total duration of one operational cycle (24 h in this study); $P_a(t)$ is aggregated output of the system at moment t ; $P_w(t)$ is wind power output at moment t , $P_{pv}(t)$ is photovoltaic power output at moment t , $P_s(t)$ is the power output of the hydropower at moment t , P_{ps} is the generation or pumping power of the pumped storage unit at moment t , P_d is the discharging power of electric energy storage at moment t . The operational cycle is defined as one day, divided into 96 intervals with a 15-min resolution.

3.3.2. Constraints

- 1) The output constraints of wind power

$$0 \leq P_w(t) \leq P_{wmax}(t) \quad (2)$$

where $P_{wmax}(t)$ is maximum wind power output at moment t .

- 2) The output constraints of photovoltaic power

$$0 \leq P_{pv}(t) \leq P_{pvmax}(t) \quad (3)$$

where $P_{wmax}(t)$ is maximum photovoltaic power output at moment t .

- 3) The output constraints of hydropower

The balance constraints for upper and lower reservoirs of hydro electric power stations:

$$V_u(t) = V_u(t-1) + 3600(I_u(t) + Q_p(t) - Q_g(t) - Q_e(t))\Delta t \quad (4)$$

$$V_d(t) = V_d(t-1) + 3600(I_d(t) + Q_g(t) - Q_p(t) + Q_e(t))\Delta t \quad (5)$$

where $V_u(t)$ and $V_d(t)$ represent the storage capacity of the upper and lower warehouses at moment t , respectively; $I_u(t)$ and $I_d(t)$ represent the incoming water flow rate of the upper and lower reservoirs at moment t , respectively; $Q_g(t)$ is the power generation flow rate of PS/hydropower at moment t ; $Q_p(t)$ is the pumping flow rate of two pumping and storage units at moment t ; $Q_e(t)$ is the ecological flow of the power station at moment t ; Δt is the step size of the calculation period.

Constraints on storage capacity size

$$\begin{cases} V_{u\min} \leq V_u(t) \leq V_{u\max} \\ V_{d\min} \leq V_d(t) \leq V_{d\max} \\ V_u(1) \leq V_u^{be} \\ V_d(1) \leq V_d^{be} \end{cases} \quad (6)$$

where $V_{u\min}$ and $V_{d\min}$ are minimum storage capacities of the upper and lower reservoirs, respectively; $V_{u\max}$ and $V_{d\max}$ are maximum storage capacities of the upper and lower reservoirs, respectively; V_u^{be} and V_d^{be} are initial storage capacities of the upper and lower reservoirs, respectively.

Water Level-Storage Capacity Coupling Constraints

$$\begin{cases} V_u(t) = f_u(Z_u(t)) \\ V_d(t) = f_d(Z_d(t)) \end{cases} \quad (7)$$

where $Z_u(t)$ and $Z_d(t)$ are water levels of the upper and lower reservoirs, respectively; $f_u(\cdot)$ and $f_d(\cdot)$ are elevation-storage curves defining the relationship between reservoir capacity and water level, respectively.

Water Level Fluctuation Constraints

$$\begin{cases} |Z_u(T) - Z_u(1)| \leq \varepsilon \\ |Z_u(t) - Z_u(t-1)| \leq \omega \end{cases} \quad (8)$$

where ε is water level difference of the upper reservoir between the start and end of the planning period; ω is water level difference between adjacent scheduling periods.

The output constraints of hydro power:

$$P_{smin} \leq P_s(t) \leq P_{smax} \quad (9)$$

where P_{smin} and P_{smax} are the minimum and maximum power outputs of the hydro unit, respectively.

Flow Constraints.

Constraints on the generation flow of conventional hydroelectric units

$$Q_{gmin} \leq Q_g \leq Q_{gmax} \quad (10)$$

where Q_g is the discharge at moment t ; Q_{gmin} and Q_{gmax} are the minimum and maximum discharge, respectively.

- 4) PS constraints

Pumped storage unit power output constraints

$$P_{psmin} \leq P_{ps} \leq P_{psmax} \quad (11)$$

where P_{psmin} and P_{psmax} are the minimum and maximum generation or pumping power, respectively.

PS power generation flow constraints

$$Q_{pmin} \leq Q_p(t) \leq Q_{pmax} \quad (12)$$

where $Q_p(t)$ is the instantaneous pumped storage PS flow, and Q_{pmin} and Q_{pmax} are the minimum and maximum pumped storage power generation flow, respectively.

Generation head constraints:

$$H_{it} = \frac{Z_u(t) - Z_u(t-1)}{2} - \frac{Z_d(t) - Z_d(t-1)}{2} \quad (13)$$

where H_{it} is the generation head of hydropower or pumped storage at moment t .

Constraints on unit output characteristic relationships

$$\begin{cases} P_s(t) = f_{1,nhq}(Q_g(t), H_i(t)) \\ P_{ps}(t) = f_{2,nhq}(Q_p(t), H_i(t)) \end{cases} \quad (14)$$

where $f_{1,nhq}$ and $f_{2,nhq}$ represent functions of the relationship between output, flow rate and head for hydropower and pumped storage, respectively.

State of charge (SOC) constraints:

The remaining water volume in the upper reservoir can be analogized to the SOC of a battery, mathematically defined as:

$$SOC_P(t) = \frac{V_u(t)}{V_{\max}} \quad (15)$$

where $SOC_P(t)$ is the remaining available water volume percentage at moment t .

5) Ramp Rate Constraints

$$\begin{cases} -\Delta P_s \leq P_s(t+1) - P_s(t) \leq \Delta P_s \\ -\Delta P_{ps} \leq P_{ps}(t+1) - P_{ps}(t) \leq \Delta P_{ps} \end{cases} \quad (16)$$

where ΔP_s and ΔP_{ps} are the maximum ramp rates of the hydro unit and pumped storage unit, respectively.

6) constraints of BES

SOC Constraints:

$$SOC(t) = SOC(t-1) + \begin{cases} \frac{\int P_c(t) \times \eta_c}{E} \\ \frac{\int P_d(t) \times \eta_d}{E} \end{cases} \quad (17)$$

$$SOC_{\min} \leq SOC(t) \leq SOC_{\max} \quad (18)$$

where $SOC(t)$ is the state of charge (SOC) of the energy storage system at moment t ; P_d is the discharging power of electric energy storage at moment t ; η_c and η_d are the charging/discharging efficiency of the energy storage system at moment t , respectively; E is the BES capacity; $SOC_{\min}$ and $SOC_{\max}$ are the lower and upper SOC limits of the system, respectively.

BES charge/discharge power constraints

$$\begin{cases} 0 \leq P_c(t) \leq \gamma_c(t)P_{\max} \\ 0 \leq P_d(t) \leq \gamma_d(t)P_{\max} \\ \gamma_c(t) + \gamma_d(t) = 1 \\ \gamma_c(t), \gamma_d(t) \in \{0, 1\} \end{cases} \quad (19)$$

where $P_{\max}$ and $P_{\max}$ are maximum charging and discharging power of the system, respectively; $\gamma_c(t)$ and $\gamma_d(t)$ charge/discharge status variables of the system at moment t , taking values 0 or 1.

7) System power balance constraints

$$P_w(t) + P_{pv}(t) + P_s(t) + P_{ps}(t) + P_d(t) - P_c(t) = P_{\text{load}}(t) \quad (20)$$

where $P_{\text{load}}(t)$ is electric load power.

3.3.3. Model solving

The optimal scheduling model for fluctuation smoothing scenarios is a high-dimensional nonlinear model with complex constraints. It is difficult to solve this model directly. Therefore, the complex scheduling model is transformed into a MILP problem [43] via a linear transformation and solved by Gurobi, a commercial solver in Matlab.

According to (14), it can be seen that there is a nonlinear relationship between the output of the unit, flow rate and head. The triangular weight method is used to linearize it. Firstly, the flow and head are divided into two intervals, $n1-1$ and $n2-1$, which can be expressed as equation (21)

$$\begin{cases} Q_{\min} = Q_0 < Q_1 < Q_2 < \dots < Q_n < \dots < Q_{n1-1} \\ H_{\min} = H_0 < H_1 < H_2 < \dots < H_n < \dots < H_{n2-1} \end{cases} \quad (21)$$

Q_n and H_n are the endpoint values for each interval of flow and level, respectively. For any range from n to $n+1$, and can be expressed in equation (22).

$$\begin{cases} Q = \alpha_0 Q_n + \alpha_1 Q_{n+1} \\ H = \delta_0 H_n + \delta_1 H_{n+1} \end{cases} \quad (22)$$

where α_0 , α_1 , δ_0 , δ_1 are all positive real numbers.

The relationship between the output of unit P, the flow rate and the head can be expressed as (23), where the output of unit P can be expressed from nonlinear to linear in terms of a and b . The output of unit P can be expressed in terms of λ_{ij} and p_{ij} .

$$\begin{cases} p_{ij} = H_i Q_j \\ P = \sum_{i=1}^{n1} \sum_{j=1}^{n2} \lambda_{ij} p_{ij} \end{cases} \quad (23)$$

where p_{ij} is the intermediate variable; λ_{ij} is the combination coefficient. The linearization of the other equations is cited in Refs. [44–46] and will not be repeated here.

3.4. CTCM-based optimization control for the WPHHS in grid ancillary services

After meeting load demand and smoothing out the system output fluctuations, the WPHHS typically retains surplus capacity, including idle BES capacity, output margins of PS, and generation flexibility of hydroelectric units (treated equivalently to PS in this study). To fully utilize these underused resources and maximize revenue, a multi-scenario utilization model is developed for deploying surplus system capacity in ancillary services. System operators can leverage this capacity to participate in electricity market transactions (e.g., frequency regulation, peak shaving, and energy arbitrage), generating additional profits through flexible dispatch of storage and hydro resources.

3.4.1. Objective function

1 The objective function is formulated as follows:

$$\max F_2 = C_{ab} + C_{fr} + C_{pr} \quad (24)$$

where F_2 denotes the total revenue; C_{ab} represents the revenue from peak-valley arbitrage, and C_{fr} and C_{pr} denote the revenues from participating in frequency regulation and peak shaving services, respectively.

$$C_{ab} = \sum_{t=1}^T (y_g(t)p_g(t) - y_p(t)p_p(t) + y_d(t)p_{d1}(t) - y_c(t)p_{c1}(t)) \quad (25)$$

where $y_g(t)$ and $y_p(t)$ denote the electricity prices for unit generation and pumping consumption at moment t ; $p_g(t)$ and $p_p(t)$ represent the power outputs of generation and pumping for PS at moment t , respectively; $y_d(t)$ and $y_c(t)$ denote the electricity prices for discharging and charging of BES at moment t ; $p_{d1}(t)$ and $p_{c1}(t)$ represent the power outputs of discharging and charging for BES at moment t .

$$C_{fr} = \sum_{t=1}^T (y_{s1}p_{s2}(t) + y_e p_e(t)) \quad (26)$$

where y_s and y_e denote the frequency regulation capacity prices of PS and battery storage, respectively; $p_{s2}(t)$ and $p_e(t)$ represent the allocated frequency regulation power of PS and battery storage at moment t , respectively.

$$C_{pr} = \sum_{t=1}^T (y_{s1}p_{s3}(t) + y_{e1}p_{e1}(t)) \quad (27)$$

where y_{s1} and y_{e1} denote the peak shaving prices of PS and BES, respectively; $p_{s3}(t)$ and $p_{e1}(t)$ represent the allocated peak shaving power of PS and BES at moment t , respectively.

2 .

3.4.2. Constraints

1) Frequency regulation-related constraints

$$\begin{cases} 0 \leq p_{s2}(t) \leq \sigma_1 p_{1 \max}(t) \\ 0 \leq p_{e1}(t) \leq \sigma_2 p_{2 \max}(t) \end{cases} \quad (28)$$

where σ_1 and σ_2 denote the maximum percentage of rated power allocated to frequency regulation capacity for PS and BES, respectively, and $p_{1 \max}$ and $p_{2 \max}$ represent the maximum allowable power output of PS and BES at moment t , respectively.

2) Peak shaving-related constraints

$$\begin{cases} 0 \leq p_{s3}(t) \leq \zeta_1 p_{1 \max}(t) \\ 0 \leq p_{e1}(t) \leq \zeta_2 p_{2 \max}(t) \end{cases} \quad (29)$$

where ζ_1 and ζ_2 denote the maximum percentage of rated power allocated to peak shaving for PS and BES, respectively.

Other constraints related to hydroelectric units, PS, and BES align with those defined in Section 3.4.2 and are not reiterated here.

3.4.3. Model solving

The CTCM is employed to solve the HESS capacity allocation model. The schematic diagram of the CTCM mechanism is illustrated in Fig. 7.

It can be seen from Fig. 7, the CTCM mimics early human tribes' settlement-seeking behavior through four core mechanisms: cooperation, competition, migration, and conflict. The optimization process of the algorithm unfolds as follows:

1) Initialization

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \dots \\ X_n \end{bmatrix} \in \mathbb{R}^{p \times d}, V = \begin{bmatrix} V_1 \\ V_2 \\ \dots \\ V_n \end{bmatrix} \in \mathbb{R}^{p \times d} \quad (30)$$

$$V_n = \begin{bmatrix} V_{n,1,1} & V_{n,1,2} & \dots & V_{n,1,d} \\ V_{n,2,1} & V_{n,2,2} & \dots & V_{n,2,d} \\ \dots & \dots & \dots & \dots \\ V_{n,m,1} & V_{n,m,2} & \dots & V_{n,m,d} \end{bmatrix} \in \mathbb{R}^{m \times d} \quad (31)$$

where X denotes the initial position of the entire society; V represents

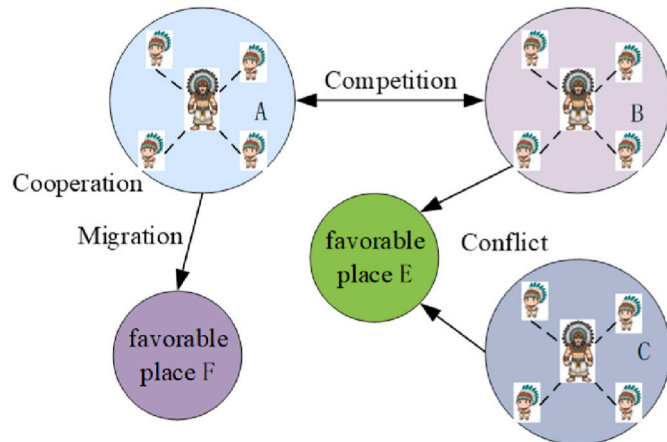


Fig. 7. Principle of CTCM algorithm.

the velocity matrix of the society; p indicates the population size in the primitive society; n signifies the number of tribes; m corresponds to the number of members per tribe; d denotes the dimensionality of the solution space.

2) Cooperation of members (exploitation)

Within a tribe, the chief manages and strategizes the tribe's future development. While most members adhere to the chief's plans and directives, each member retains individual autonomy, leading to fluctuating loyalty over time. This loyalty exhibits chaotic behavior as temporal iterations progress, and the chaotic dynamics become more pronounced as the number of tribes increases. To model this phenomenon, the concept of a sine chaotic map is introduced to characterize the evolution of member loyalty.

$$r_t = \begin{cases} U(0, 1), & \text{if } t = km, k \in \mathbb{Z} \\ \sin\left(\frac{\pi}{2} r_t\right), & \text{else} \end{cases} \quad (32)$$

where r_t denotes the loyalty of an individual member.

Each member consults the tribal chief to receive directives for their next action. While adhering to these directives, the member incorporates its own experiential knowledge, which supplements decision-making with a stochastic element to diversify potential actions.

$$V_{n,m,d}^{t+1} = \frac{3}{5} V_{n,m,d}^t + c_1 r_{1,n,m,d}^t (X_{n,m,d}^{\text{best}} - X_{n,m,d}^t) + c_2 r_{2,n,m,d}^t (X_{\text{tribe}}^{\text{best}} - X_{n,m,d}^t) \quad (33)$$

where $V_{n,m,d}^{t+1}$ and $V_{n,m,d}^t$ denote the velocities of the m -th member in the n -th tribe at time $t+1$ and t , respectively; $X_{n,m,d}^{\text{best}}$ represents the best fitness position of the member across the entire optimization horizon; $X_{n,m,d}^t$ signifies the position of the member at time t ; $X_{\text{tribe}}^{\text{best}}$ denotes the best fitness position of the n -th tribe; c_1 and c_2 correspond to the weights for experiential adherence and chief compliance of the tribe member, respectively; and $r_{1,n,m,d}^t$ and $r_{2,n,m,d}^t$ represent the chaotic loyalty coefficients of individual members, governed by the sine chaotic map dynamics.

3) Competition of tribes (exploration)

Stochastic inter-tribal conflicts occur during iterations, where weaker tribes (those with lower fitness values) are pruned or displaced, while stronger tribes (higher fitness) remain unaffected and continue dominating resource allocation.

$$R_{n,m,d}^t = c_3 r_{3,n,m,d}^t (X_{n,m,d}^t - X_{\text{rival}}^{\text{best}}) \quad (34)$$

$$V_{n,m,d}^{t+1} = \begin{cases} V_{n,m,d}^{t+1} - R_{n,m,d}^t, & \text{if } F_n^{\text{best}} < F_{\text{rival}}^{\text{best}} \\ V_{n,m,d}^{t+1}, & \text{else} \end{cases}$$

where $X_{\text{rival}}^{\text{best}}$ denotes the best position discovered by opposing tribes; F_n^{best} represents the best fitness value of the n -th tribe; $F_{\text{rival}}^{\text{best}}$ signifies the best fitness value of opposing tribes; c_3 corresponds to the tribal escape coefficient governing retreat dynamics during conflicts; $r_{3,n,m,d}^t$ denotes the chaotic stochastic factor modulating retreat velocity, derived from sine chaotic map iterations. The remaining capacity allocation process based on the CTCM algorithm [47] is illustrated in Fig. 8.

The specific solution steps are as follows:

- 1) Retrieve the remaining capacity and state constraints of system components at each time step from the control model;
- 2) Determine the capacity allocation scheme with the objective of maximizing revenue;

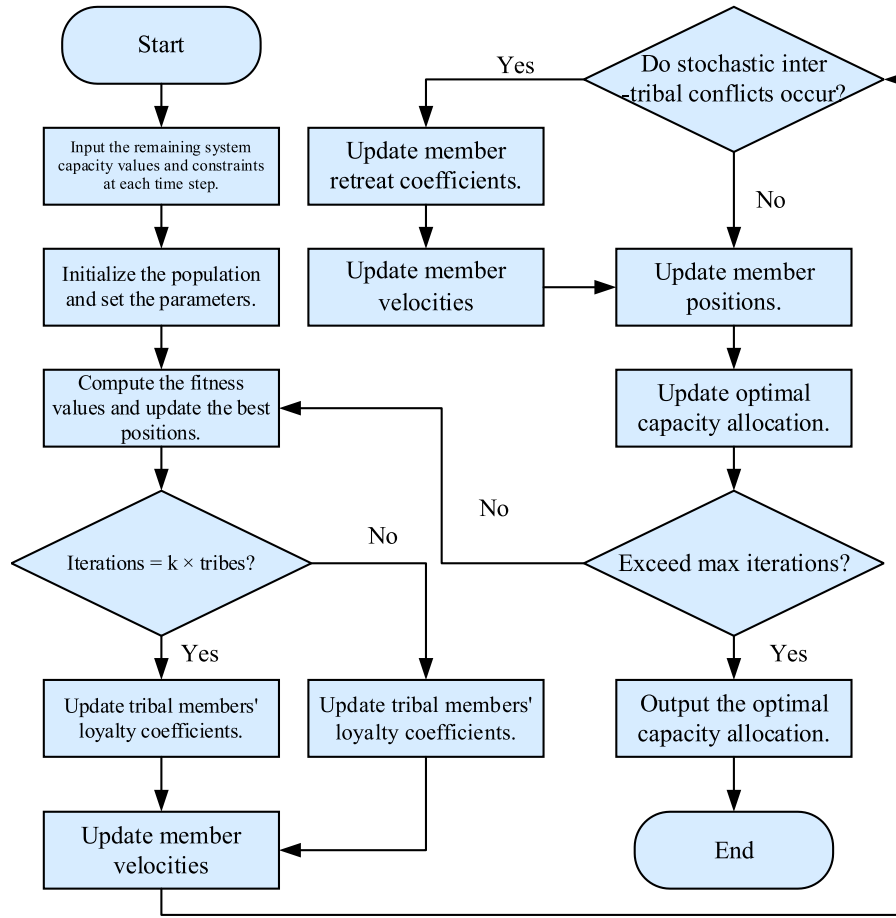


Fig. 8. The process of HESS capacity allocation based on CTCM.

- 3) Initialize the population size and maximum iterations for the algorithm;
- 4) Obtain the optimal allocation results via iterative optimization using the CTCM algorithm.

3.4.4. Cost-benefit analysis

1. System annualized net income

$$\begin{cases} R_{\text{net}} = R_{\text{tot}} - C_{\text{opex}} \\ C_{\text{opex}} = C_{\text{fix}} + C_{\text{var}}^{\text{bes}} + C_{\text{var}}^{\text{phs}} + C_{\text{deg}}^{\text{bes}} \end{cases} \quad (35)$$

Where R_{net} is the system annualized net income, R_{tot} is the annualized total income and C_{opex} is the annualized operation and maintenance cost. C_{fix} , $C_{\text{var}}^{\text{bes}}$, $C_{\text{var}}^{\text{phs}}$, and $C_{\text{deg}}^{\text{bes}}$ are fixed operation and maintenance costs, battery variable operation and maintenance costs, PS variable operation and maintenance costs, and battery degradation costs, respectively.

2. Return on investment and payback period

$$\begin{cases} ROI = \frac{R_{\text{net}}}{C_{\text{capex}}} \times 100\% \\ T_p = \frac{C_{\text{capex}}}{R_{\text{net}}} \\ C_{\text{capex}} = C_{\text{bes}} + C_{\text{phs}} \end{cases} \quad (36)$$

Where ROI and T_p represent the return on investment and payback period, respectively; C_{capex} is the capital expenditures, C_{bes} and C_{phs} are the capital costs of BES and PS, respectively.

The overall logical structure of this paper is illustrated in Fig. 9, with the specific workflow as follows: First, uncertainty optimization processing for wind and solar energy is conducted at the input layer. Using the LHS method, a large number of wind and solar scenarios are generated. Subsequently, the L-ISODATA is employed to select typical scenarios. Concurrently, quantitative analysis of DBI and DI is performed to ensure the representativeness and rationality of the selected typical scenarios. Second, the optimization layer constructs a two-tier optimization strategy for multi-energy complementary systems across multiple timescales: the upper layer minimizes system fluctuation handling volume as its core objective, solved using the MILP approach; the lower layer maximizes ancillary service revenue as its key objective, solved using the CTCM algorithm. Finally, the output layer conducts phased validation and assessment: First, the foundational effectiveness of the optimization strategy is verified through system volatility validation and ancillary service revenue validation. Subsequently, comprehensive stability and overall performance under various operating conditions are evaluated through multi-scenario robustness validation and multi-dimensional performance assessment.

4. Case study

4.1. Data sources and parameter settings

The data is sourced from a domestic power plant with 950 MW wind and 910 MW photovoltaic capacities. The hybrid pumped storage plant comprises two reservoirs and one pumping station, with data recorded at 15-min resolution. The water level difference between the upper and lower reservoirs is maintained within $\varepsilon = 0.25$ m, and the inter-temporal water level fluctuation is constrained below $\omega = 0.1$ m to prevent

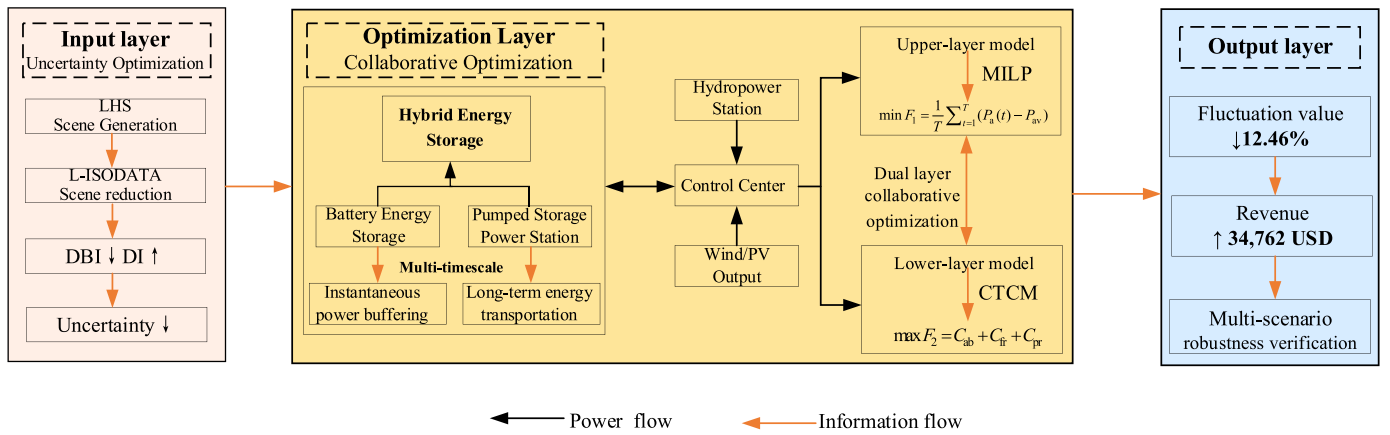


Fig. 9. A dual-layer optimization framework and validation system for multi-time-scale WPHHS.

destabilizing effects on the coastal slope [48]. Parameters for hydropower and PS are listed in Tables 2 and 3.

The energy storage system employs lithium iron phosphate batteries configured with 50 MW/100 MWh capacity and charge/discharge efficiency of 0.95. The SOC limits for the battery and PS are set to [0.1 0.9] and [0.1 1], respectively. The unit cost of BES is USD 245/kW. PS utilizes existing power plant capacity expansion and retrofitting, with a converted cost of approximately USD 50/kW. Operating and maintenance costs are set at 1.5 % of total investment per annum. The payback period is 15 years, with a discount rate of 8 % [49,50]. The time-of-use (TOU) electricity price, frequency regulation capacity price, and peak shaving capacity price are detailed in Table 4.

4.2. Comparison of system output smoothing effects under different schemes

In order to clearly analyze the specific role of the WPHHS in smoothing out volatility, we can start our analysis by comparing the following two systems:

Wind-PV-hydro system (WPHS): The WPHS represents a conventional wind-PV-hydro control approach that coordinates wind energy, PV power generation, and hydropower without incorporating any energy storage systems. In this system, electricity generated from wind, PV, and hydropower is directly used to meet load demand. When generation is insufficient, power is purchased from the external grid to bridge the gap.

Wind-PV-hydro system with pumped storage system (WPHPS): The WPHPS extends the WPHS by integrating a pumped storage system, which regulates energy supply through water pumping and discharging.

The model is solved using Gurobi, with the system fluctuation mitigation performance of different dispatch systems illustrated in Fig. 10 and Table 5.

It can be seen from Fig. 9 and Table 5, the WPHS exhibits a system output fluctuation of 81.60 MW, with wind and photovoltaic curtailment rates of 21.3 % and 18.95 %, respectively, attributed to the lack of energy storage. In the WPHPS, the integration of PS reduces the fluctuation to 72.73 MW and lowers wind and photovoltaic curtailment

Table 2
Characteristic parameters of hydropower and PS.

Type	Maximum power generation (MW)	Maximum pumping power (MW)	Maximum flow (m ³ /s)	Climb rate (MW/15min)
Hydropower	440	/	630	70
PS	400	360	585	70

Table 3
Characteristic parameters of hydropower and pumped storage.

Reservoir	Normal Water level(m)	Minimum water level(m)	Initial water Level(m)	Inflow rate (m ³ /s)
Upper	1105	1059	1100	60
Lower	956	920	959	0

Table 4
Time of use electricity price, peak shaving and frequency regulation capacity price.

Type	Time period	PS	BES
Peak price period (USD/MWh)	8:00–12:00; 17:00–21:00	109	109
Valley price period (USD/MWh)	0:00–8:00	46	46
Normal price period (USD/MWh)	12:00–17:00; 21:00–24:00	87	87
Peak shaving (USD/MWh)	/	7	10
Frequency regulation (USD/MWh)	/	17	24

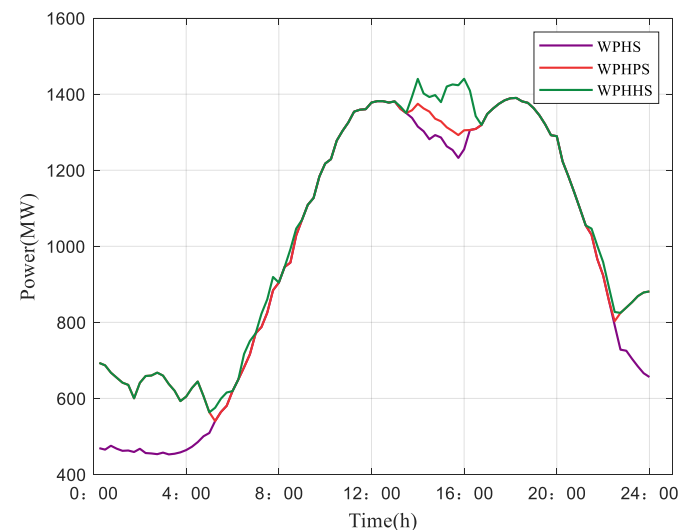


Fig. 10. Comparison of calming effects under different systems.

rates to 4.46 % and 10.05 %, thereby improving energy utilization. The WPHHS, which combines pumped storage and battery ESS, exhibits the lowest fluctuation (71.43 MW) compared to the two aforementioned systems, it achieves reductions of 12.46 % and 1.78 %, respectively. And it eliminates curtailment (0 % for both wind and PV). These results

Table 5
Characteristic parameters of different models.

System	Fluctuation (MW)	Wind Curtailment Rate (%)	PV Curtailment Rate (%)
WPHS	81.60	21.23	18.95
WPHPS	72.73	4.46	10.05
WPHHS	71.43	0	0

demonstrate that integrating pumped storage and BES enhances energy utilization and strengthens the system's coordinated control capabilities.

Fig. 11 shows the output curves of the WPHHS.

It can be seen in Fig. 11, during the 00:00–05:00 period with low load demand and high wind power output, the surplus wind power is primarily absorbed by the PS through pumping operations. In the 05:00–13:45 and 17:00–22:15 periods, when load demand exceeds renewable generation, the hydropower and PS (in generation mode) compensate for the energy deficit, while the battery storage collaboratively mitigates system output fluctuations. During 13:45–17:00, renewable generation continuously exceeds load demand, with the surplus energy absorbed by both pumped storage pumping and battery charging. Thus, the WPHHS significantly reduces output fluctuations, improves energy utilization, and eliminates wind and photovoltaic curtailment while reliably meeting base load requirements.

Fig. 12 illustrates the water level variation curve of the upper reservoir in the hydropower station during system fluctuation mitigation using the WPHHS.

It can be seen in Fig. 12 that during 00:00–05:00 period, the PS operate in pumping mode, raising the water level of the upper reservoir. From 05:00–13:45 and 17:00–22:15, when load demand exceeds wind and photovoltaic generation, the hydro units increase water discharge to meet energy deficits, resulting in a decline in the upper reservoir level. Between 13:45–17:00, reduced load demand triggers pumping operations by the PS, elevating the upper reservoir level to enhance generation efficiency. In the 22:00–24:00 period, as load demand diminishes, the hydro units resume pumping, causing the upper reservoir level to rise. The total water level decline ($0.22 \text{ m} < 0.25 \text{ m}$, the safety threshold) remains within operational limits, ensuring system stability.

4.3. Comparative analysis of smoothing effects under different hydrological conditions

To investigate the fluctuation mitigation performance of the hybrid complementary system under different hydrological conditions, a

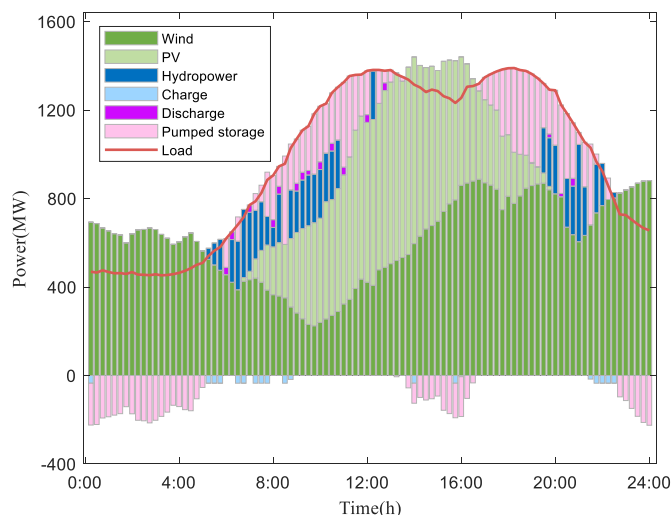


Fig. 11. System output curve of the WPHHS.

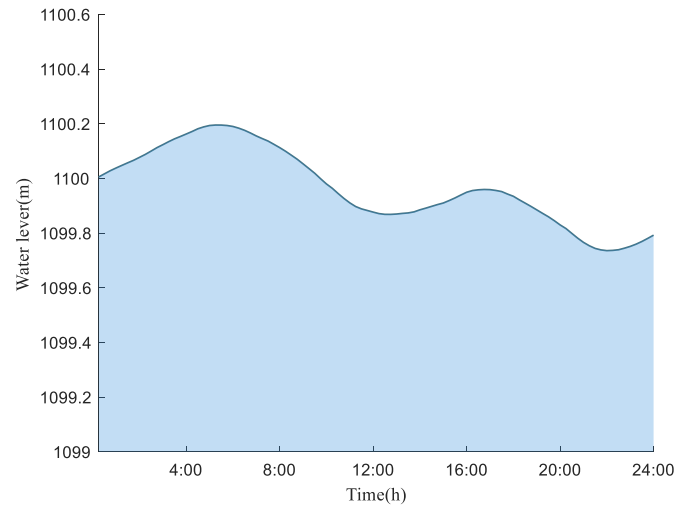


Fig. 12. Changes in water level of the upper reservoir.

scenario-based analysis is conducted for the hydropower plant during wet, normal, and dry seasons, with the smoothing effectiveness summarized in Table 6.

It can be seen in Table 6 that the integration of pumped storage and BES systems significantly reduces system output fluctuations across all hydrological seasons, with the WPHHS achieving 12.46 %, 16.05 %, and 20.61 % reductions in output fluctuations during wet, normal, and dry seasons compared to the WPHPS, respectively, where the dry season exhibits the most notable improvement. From the overall trend, the volatility of the system shows a decreasing law with the change of hydrological period: it is the largest during the Wet season, the second largest during the normal season, and the smallest during the dry season. This phenomenon mainly stems from the difference in the peak adjustment ability of hydropower systems in different hydrological conditions: in the wet season, hydropower mainly undertakes base load operation, and its peak adjustment ability and regulation ability for wind power generation are relatively weak; while in the dry season, hydropower systems have the optimal regulation performance, and thus the system fluctuation is the smallest. This correlation between hydrological season and system volatility reflects the important role of hydropower regulation capability in power system stability.

4.4. Economic comparison for the WPHHS under different operational strategies

To quantify the comprehensive benefits of utilizing surplus capacity for ancillary services, this study designs coordinated control strategies between PS and the BES using a controlled variable method, as follows:

Strategy 1: PS prioritize peak-valley arbitrage; Battery storage focuses on peak shaving and frequency regulation.

Strategy 2: PS prioritizes peak-valley arbitrage; BES operates in multi-scenario mode (simultaneous arbitrage, peak shaving, and frequency regulation).

Strategy 3: PS and BES operate in multi-scenario mode.

Strategy 4: PS adopts multi-scenario operations; BES prioritizes peak shaving and frequency regulation.

Table 6

The effect of stabilizing system fluctuations under different hydrological conditions.

Systems	Wet season (MW)	Normal season (MW)	Dry season (MW)
WPHS	81.60	69.95	58.61
WPHPS	72.73	60.23	47.87
WPHHS	71.43	58.72	46.53

The CTCM optimization algorithm is applied to solve the model, with the economic performance comparisons across strategies summarized in Table 7.

It can be seen in Table 7, Strategy 4 achieves the highest system revenue at 34,762 USD, while Strategy 3 yields the lowest revenue at 28,770 USD, which is 17.23 % lower than Strategy 4. Strategy 2 ranks second with a revenue of 33,084 USD, trailing Strategy 4 by 4.83 %. This variation results from the distinct technical advantages of BES and PS: BES excels in fast response and high energy conversion efficiency, maximizing revenue when fully allocated to frequency regulation services, whereas PS leverage long-duration energy storage capabilities to dominate peak-valley arbitrage while partially participating in peak shaving and frequency regulation to enhance earnings.

Under Strategy 4, the CTCM capacity allocations and revenue distributions for PS and BES are detailed in Fig. 13 and Table 8.

It can be seen in Fig. 13 and Table 8, BES prioritizes frequency regulation services due to their highest revenue contribution (77.81 % of total system revenue), allocating 637 MW of capacity to frequency regulation and the remaining capacity to peak shaving. For PS, simultaneous participation in three ancillary services maximizes revenue: 946 MW for peak-valley arbitrage, 835 MW for peak shaving, and 803 MW for frequency regulation. The revenue distribution is dominated by frequency regulation (50.72 %), followed by peak-valley arbitrage (29.54 %) and peak shaving (19.74 %). This allocation reflects the lower frequency regulation revenue of PS compared to battery storage, making full capacity dedication to frequency regulation suboptimal, and the need to balance volatile arbitrage gains (due to fluctuating peak-valley electricity prices) with other services to mitigate risks. Overall, Strategy 4 effectively reduces idle capacity by deploying 1,033 MW for peak shaving, 960 MW for frequency regulation, and 946 MW for arbitrage, demonstrating synergistic utilization of PS and BES.

To more balancedly assess the economic feasibility of hybrid energy storage systems, this section further introduces capital expenditure (CAPEX) and operation and maintenance expenditure (OPEX), and uses the return on investment (ROI) and payback period indicators for an expanded analysis. The results are shown in Table 9.

As shown in Table 9, after accounting for CAPEX and OPEX, all four strategies achieve positive returns, with payback periods of less than 8 years, significantly shorter than the 15-year economic lifecycle of a hybrid energy storage system. Strategy 4 performs best, achieving an ROI of 15.3 % and a payback period of 6.1 years. This not only increases system revenue (an additional \$34,762 compared to the baseline) but also demonstrates strong economic viability. Strategies 1–3, with ROIs of 9.2 %, 11.4 %, and 12.6 %, respectively, and payback periods ranging from 6.9 to 7.8 years, also demonstrate considerable investment appeal.

Time-of-use pricing serves as a crucial mechanism for enhancing the economic efficiency and reliability of power systems. To investigate the impact of electricity pricing on system benefits, this study employs multiple pricing schemes for sensitivity analysis: while maintaining the benchmark off-peak rate unchanged, the peak-to-off-peak price ratio is set within the range of 1–7 [51]. The comparative results are presented in Fig. 14.

It can be seen from Fig. 14, as the peak-to-valley ratio increases, both ancillary service revenue and ROI show a significant upward trend, indicating that the system's economic benefits are effectively improved despite significant electricity price fluctuations. Specifically, as the peak-to-valley ratio increases from 1 to 7, ancillary service revenue increases from approximately \$22,100 to \$66,300, while ROI increases

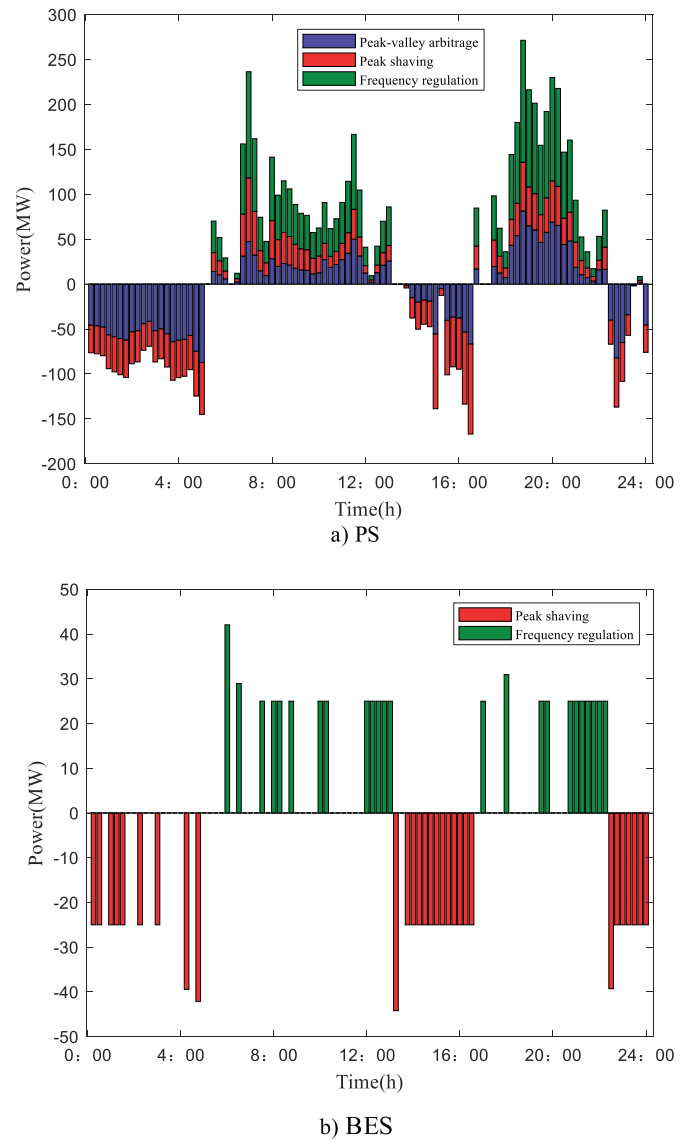


Fig. 13. Capacity allocation results based on CTCM.

from 9.7 % to 19.7 %. This result demonstrates that a higher peak-to-valley price difference can help improve system profitability and return on investment, but it also suggests that the impact of electricity price fluctuations on system operations must be comprehensively considered during implementation.

4.5. Economic comparison across optimization algorithms

To validate the effectiveness and superiority of the CTCM algorithm, the model is solved using three benchmark optimization algorithms: Particle Swarm Optimization (PSO), Starfish Optimization Algorithm (SFOA), and Sparrow Search Algorithm (SSA). The parameters are set with a population size of 60 and 500 iterations. Taking the revenue allocation of PS as an example, Fig. 15 compares the convergence curves of different algorithms. The results obtained after conducting 30 simulations for each of the four algorithms are shown in Table 10. To validate the algorithm's scalability, model optimization verification was conducted sequentially over 12-h, 24-h, and 48-h periods. The results are presented in Table 11.

It can be seen in Fig. 15 and Table 10, the CTCM algorithm demonstrates significant advantages across all performance metrics. Its convergence curve exhibits rapid and stable descent, far outperforming

Table 7
Economic performance comparison across operational strategies.

Economic/USD	Strategy 1	Strategy 2	Strategy 3	Strategy 4
PS	21882	26306	21882	26306
BES	8456	6888	6888	8456
Total revenue	30338	33194	28770	34762

Table 8
Capacity allocation and revenue situation under Strategy 4.

Type	Peak shaving		Frequency regulation		Peak-valley arbitrage	
	Capacity/MW	Revenue/USD	Capacity/MW	Revenue/USD	Capacity/MW	Revenue/USD
PS	835	5194	803	13342	946	7770
BES	198	1876	157	6580	/	/
Total	1033	7070	960	19922	946	7770

Table 9
Return on investment and payback period for different strategies.

Strategy	Annualized income/ 10^4 USD	ROI/%	Payback period/years
Strategy 1	698.57	9.2	7.8
Strategy 2	734.29	11.4	7.2
Strategy 3	755.71	12.6	6.9
Strategy 4	810.86	15.3	6.1

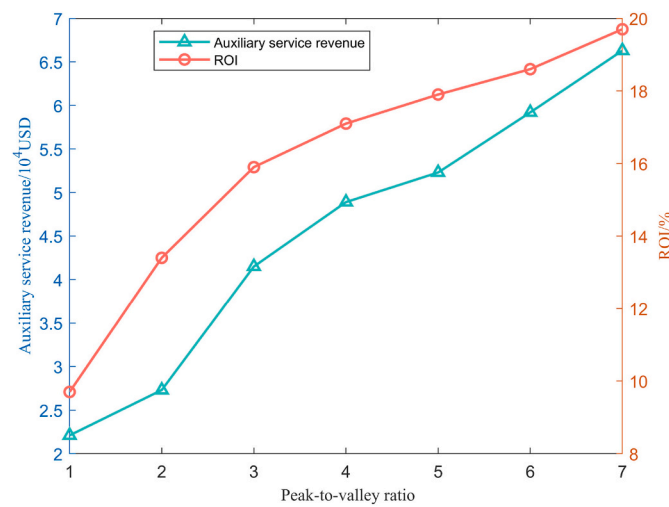


Fig. 14. Sensitivity analysis results for peak-to-valley ratio in time-of-use electricity pricing.

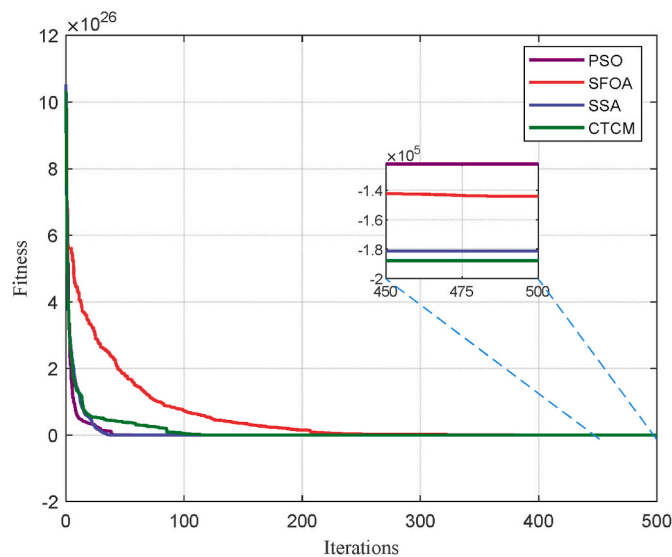


Fig. 15. Algorithm convergence curves.

Table 10
Comparison of results from four algorithms.

Algorithm	Average value	Optimal value	Worst value	CV	Average time/s
PSO	-122,125	-143,434	-111,450	0.85	4.7
SFOA	-144,141	-147,221	-129,634	1.10	10.3
SSA	-181,427	-190,245	-175,334	1.42	6.7
CTCM	-187,928	-198,763	-179,445	0.21	5.8

Table 11
Scalability comparison of four algorithms.

Algorithm	$48 \times 3(12h)$		$96 \times 3(24h)$		$192 \times 3(48h)$	
	Time/s	Fitness	Time/s	Fitness	Time/s	Fitness
PSO	3.4	-87,510	4.7	-122,125	/	/
SFOA	7.2	-90,843	10.3	-144,141	13.8	-298,741
SSA	4.9	-90,821	6.7	-181,427	9.5	-334,657
CTCM	4.5	-92,517	5.8	-187,928	7.7	-387,620

other algorithms. Regarding fitness values, CTCM ranks first in average ($-187,928$), optimal ($-198,763$), and worst ($-179,445$) values. Its average computation time (5.8s) ranks second only to PSO. Moreover, it exhibits the lowest coefficient of variation(CV) at 0.21, indicating greater stability, reduced susceptibility to random factors, and superior robustness and reliability. As shown in Table 11, as problem scale increases, CTCM consistently maintains the optimal fitness value ($-387,620$) while requiring less computational time than comparison algorithms, demonstrating its exceptional scalability, stability, and efficiency.

The SOC curves of the BES and PS during the control period is shown in Fig. 16.

It can be seen in Fig. 16, from 0:00 to 5:00 and 14:00 to 18:45, PS is responsible for energy transfer and absorbs excess wind energy; from 10:00 to 12:00 and 18:45 to 20:45, HESS synergistically suppresses system fluctuations. And the SOC of the BES remains strictly within the

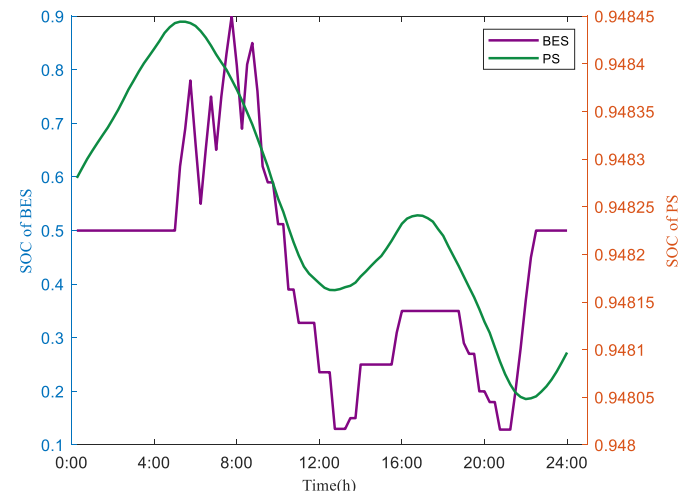


Fig. 16. The curves of the SOC.

safe operating range of 0.1–0.9 throughout the control period, while the PS exhibit minimal SOC fluctuations around a baseline of 0.984 (within the 0.1–1.0 safety margin), with no overcharge or over-discharge events observed for either technology. This demonstrates excellent operational stability and safety for both storage systems under complex multi-scenario coupled conditions. Notably, the BES maintains an initial SOC near 0.5 at the start of each control cycle, whereas the PS sustain a high SOC baseline of 0.948, reflecting divergent SOC regulation strategies that provide optimal initial energy reserve conditions for subsequent operational cycles.

4.6. Multi-scenario robustness verification and performance evaluation

To further evaluate the robustness of the multi-energy hybrid system in practical applications, the paper introduced four extreme scenario analyses. These scenarios simulate the system's responsiveness and adaptability to external shocks.

Case A. Considering an extreme weather scenario, a typhoon or sandstorm causes a 50 % drop in wind and solar output to test the system's stability under severe fluctuations in renewable energy.

Case B. Simulating a combination of dry years and abundant solar years, the paper explored the synergistic effects of hydropower and wind and solar complementarity and the adaptability of the multi-source energy scheduling strategy.

Case C. Considering the uncertainty of the electricity market, the paper introduced a market volatility scenario in which the peak-to-valley price difference was tripled to evaluate the system's economic feasibility and scheduling strategy under severe price fluctuations.

Case D. To simulate the impact of equipment failure on the system, the paper studied an equipment failure scenario, assuming a temporary 40 % reduction in BES capacity, to analyze the system's emergency response and capacity allocation capabilities in the event of an equipment failure.

Case E. Baseline scenario.

The specific impacts of these extreme scenarios on system performance are shown in Table 12.

It can be seen in Table 12, the proposed model maintains optimal convergence under extreme scenarios. However, the system fluctuation suppression rate drops to 8.2 % in the typhoon scenario due to the sudden drop in wind and solar output, which prevents the PS from fully storing energy. Profits increase to \$41,586 under the electricity price fluctuation scenario, demonstrating the model's dynamic response to market signals. Although the fluctuation suppression rate decreases in the BES failure scenario, the CTCM reallocation of PS capacity maintains profits above 85 % of the baseline level, highlighting the strategy's resilience. Overall, the system demonstrates strong robustness and maintains high stability under various extreme scenarios.

In order to comprehensively evaluate the feasibility and application

Table 12

System performance changes under different extreme scenarios.

Case	Systematic volatility changes/%	Power curtailment rate/%	Changes in earnings/USD
Case A	−8.2	4.3	+28,115
Case B	−10.1	1.8	+31,240
Case C	−13.9	0	+41,586
Case D	−9.7	2.5	+29,874
Case E	−12.46	0	+34,762

effect of this method, this article selected four typical power grid regions for analysis, including data from Northwest and Southwest China [52], the European Union [53], and North America [54]. Table 13 shows the optimized dispatch results of different power grid regions.

It can be seen in Table 13, the proposed method effectively reduces system output fluctuations when optimized in all four regions, achieving the highest performance in Southwest China, with a magnitude of 15.47 %, and the lowest performance in North America, with a magnitude of 10.68 %. This research method demonstrates significant cross-regional applicability, effectively smoothing system output fluctuations and improving renewable energy consumption in power grids in China, the West, and other regions. A hybrid energy storage solution combining batteries and pumped storage can completely eliminate wind and solar curtailment and significantly enhance grid stability. Furthermore, this strategy can generate substantial economic benefits across all regions by participating in ancillary services such as frequency regulation and peak-valley arbitrage. Multi-regional validation results demonstrate the method's excellent scalability and adaptability, enabling it to flexibly adapt to diverse grid structures and resource conditions, demonstrating its broad potential for widespread adoption.

5. Conclusion and future work

5.1. Conclusion

The WPHHS has great application prospects in dealing with wind and photovoltaic fluctuations and alleviating the problem of wind and photovoltaic power abandonment. In order to further explore its operation mode, this study established a multi-energy complementary system architecture based on "BES -PS dual-core drive", which realizes energy space-time transportation through PS and completes power instantaneous buffering through BES, forming a "double insurance" mechanism for smoothing system output fluctuations, and effectively improving the problem of energy storage idleness. The main conclusions of the paper are as follows:

- 1) LHS-generated renewable scenarios combined with L-ISODATA clustering effectively mitigate the impact of wind/PV generation uncertainty on control outcomes.
- 2) The proposed system reduces output volatility by 12.46 % and 1.78 % compared to conventional WPHS and WPHPS systems, respectively, achieving zero curtailment rates. Under varying hydrological conditions (wet, normal, and dry seasons), the system maintains optimal fluctuation mitigation, with volatility decreasing progressively (most notably in dry seasons), demonstrating robust adaptability to complex scenarios.
- 3) The capacity allocation model provides 1,033 MW of peak shaving capacity, 960 MW of frequency regulation capacity, and 946 MW for peak-valley arbitrage, reducing idle capacity while generating 34,762 USD in additional revenue.
- 4) From a cost-benefit perspective, WPHHS achieves a ROI of 15.3 % with a payback period of 6.1 years. Moreover, the higher the peak-to-valley ratio, the greater the system's ROI. Under extreme operating

Table 13

Optimization results in different regions.

Region	Systematic volatility changes/%	Power curtailment rate/%	Changes in earnings/USD
Northwest China	−12.46	0	+34,762
Southwest China	−15.47	0	+33,025
European Union	−11.30	0	+33,500
North America	−10.68	0	+34,587

conditions, simulation results for four typical extreme scenarios demonstrate that the proposed model maintains high stability and robustness, exhibiting strong interference resistance. Furthermore, compatibility tests across three distinct regional grid types reveal that the model effectively adapts to varying grid topologies and operational environments, showcasing exceptional adaptability and scalability.

In summary, the proposed model enhances system stability, renewable utilization, and economic performance through synergistic coordination of multi-energy resources, providing a reliable and flexible solution for power systems with high renewable penetration.

5.2. Future work

While effective in practical examples, this method still requires refinement in terms of robustness, error sensitivity, and physical fidelity. Multi-day stress testing (95th/99th percentiles) and rolling optimization will be conducted to address distribution shifts and extreme weather conditions. Opportunity constraints, distributed robustness, and CVaR will be used to quantify the impact of wind and solar forecast errors on power balance and SOC safety. Frequency regulation mileage fees, performance coefficients, energy bias costs, and "energy budget + SOC rebate" will be added to the model to enhance the credibility of the results.

For project implementation, market participation rules (capacity certification, minimum duration, response time, energy neutrality, and participation caps), grid connection specifications (ride-through capability, primary/fast frequency response, reactive power/power factor, and telemetry timeliness), and metering and settlement will be transformed into explicit constraints and cost items, incorporating licensing and safety compliance costs. Further consideration will be given to the inherent feedback of large-scale energy storage participation on price formation, and scaled up using cross-regional, multi-year data and decomposition algorithms to develop a deployable solution that balances technical, economic, and regulatory considerations.

CRedit authorship contribution statement

Xiaojuan Han: Writing – review & editing, Resources, Project administration, Funding acquisition, Data curation. **Xiaoyan Yang:** Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Formal analysis. **Tianyang Bai:** Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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